

Paper 1B drill: units, precision and figures

Sixteen items. The cheapest marks on the paper.

IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **24 marks**

A gradient carries the unit of the vertical axis divided by the unit of the horizontal axis. A logarithm, and anything you take a logarithm of, has no unit at all.

A *the unit of a gradient*

- A1. T^2 is plotted against L , where T is measured in seconds and L in metres. [1]
- A2. v is plotted against t , where v is measured in metres per second and t in seconds. [1]
- A3. P is plotted against T^4 , where P is measured in watts and T in kelvin. [1]
- A4. V is plotted against $1/\lambda$, where V is measured in volts and λ in metres. [2]
- A5. $\ln N$ is plotted against t , where N is a number of nuclei and t is measured in seconds. [2]

B *resolution, precision and what actually limits you*

- B1. A digital balance displays 24.36 g. State its resolution and a suitable uncertainty for the reading. [2]
- B2. A stopwatch reads to 0.01 s, but it is started and stopped by hand. State what actually limits the uncertainty, and estimate it. [2]
- B3. Following on from B2, a student times 20 complete oscillations instead of one. Explain what this does to the uncertainty in a single period, with a number. [2]
- B4. A ruler is divided in millimetres. State the uncertainty in a single reading, and then in a *length* measured with it. Explain the difference. [2]
- B5. A micrometer reads 0.03 mm when fully closed. Name this fault and state what must be done to every reading. [2]

C *significant figures*

- C1. A calculator gives 0.0471698... from data quoted to three significant figures. Write the answer down. [1]
- C2. A distance of 250 m is measured to the nearest metre. Explain why writing 250 m is ambiguous, and write it unambiguously. [2]
- C3. A speed is calculated from a distance known to four significant figures and a time known to two. State how many significant figures the speed deserves, and why. [2]

D *checking an equation*

- D1. Show that $v^2 = u^2 + 2as$ is homogeneous. [2]
- D2. A student writes $T = 2\pi\sqrt{g/L}$ for a pendulum. Use units to show that this cannot be right. [2]
- D3. Resistivity is defined by $R = \rho L/A$. Derive the unit of ρ . [2]

Answers

A *the unit of a gradient*

- A1. $\text{s}^2 \text{m}^{-1}$.
- A2. $\text{m s}^{-1}/\text{s} = \text{m s}^{-2}$, an acceleration.
- A3. W K^{-4} .
- A4. $\text{V}/\text{m}^{-1} = \text{V m}$. Dividing by a reciprocal metre multiplies by a metre, which is the step most often got backwards.
- A5. No unit. You may only take the logarithm of a pure number, so $\ln N$ is unitless, and the gradient is $1/\text{s} = \text{s}^{-1}$. (Strictly one plots $\ln(N/N_0)$ or $\ln(N/1)$; the quantity logged must be dimensionless.)
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B *resolution, precision and what limits you*

- B1. The resolution is 0.01 g, the smallest change it can display. A reasonable uncertainty is ± 0.01 g, the last digit.
- B2. Human reaction time, not the display. Starting and stopping by hand costs roughly 0.1 to 0.2 s, some twenty times the resolution, so quoting ± 0.01 s would badly overstate the precision. Use about ± 0.2 s.
- B3. The ± 0.2 s is incurred *once* for the whole run, then divided by 20 when the total is divided to find one period: $0.2/20 = \pm 0.01$ s per period. Twenty times better, for no extra equipment. This is why the instruction to time many oscillations appears in every practical.
- B4. A single reading is uncertain by about half a division, ± 0.5 mm. But a *length* is the difference between two readings, one at each end, and for a difference the absolute uncertainties add: ± 1 mm. Even with one end placed on the zero mark you have still made a judgement there.
- B5. A zero error, which is systematic. Subtract 0.03 mm from every reading. It shifts all results by the same amount and will never show up as scatter, which is why repeating the measurement cannot reveal it.
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C *significant figures*

- C1. 0.0472, three significant figures, matching the data. The leading zeros are not significant.
- C2. As written it could be two significant figures or three: the zero might be a measured digit or a placeholder. Write 2.50×10^2 m to mean three.
- C3. Two. A result can be no more precise than the weakest measurement in it, so the two-figure time sets the limit however precisely the distance is known.
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D *checking an equation*

- D1. $[v^2] = (\text{m s}^{-1})^2 = \text{m}^2 \text{s}^{-2}$, and $[u^2]$ is the same. $[2as] = (\text{m s}^{-2})(\text{m}) = \text{m}^2 \text{s}^{-2}$. Every term has the same unit, so the equation is homogeneous. (Homogeneity is a necessary check, not a sufficient one: it cannot detect a wrong pure number such as the 2.)
- D2. $\sqrt{g/L}$ has unit $\sqrt{\text{m s}^{-2}/\text{m}} = \sqrt{\text{s}^{-2}} = \text{s}^{-1}$, which is a frequency, not a time. The correct form $2\pi\sqrt{L/g}$ gives $\sqrt{\text{m}/\text{m s}^{-2}} = \sqrt{\text{s}^2} = \text{s}$. The student has inverted the fraction.
- D3. $\rho = \frac{RA}{L}$, so the unit is $\frac{\Omega \cdot \text{m}^2}{\text{m}} = \Omega \text{ m}$.